

less efficient as the temperature rises—the new process excels at higher temperatures.

Called photon-enhanced thermionic emission (PETE), the process promises to surpass the efficiency of existing photovoltaic and thermal conversion technologies. The materials needed to build a device to make the process work are cheap and easily available, meaning the power that comes from it will be affordable.

Most photovoltaic cells use the semiconducting silicon to convert the energy from photons of light into electricity. But the cells can only use a portion of the light spectrum, with the rest just generating heat. This heat from unused sunlight and inefficiencies in the cells themselves account for a loss of more than 50 per cent of the initial solar energy reaching the cell. If this wasted heat energy could somehow be harvested, solar cells could be much more efficient.

By coating a piece of semiconducting material with a thin layer of the metal cesium, the researchers made the material able to use both light and heat to generate electricity.

Breakthrough technology for advanced chip designs

Applied Materials claims to have developed the first and only film-deposition technology capable of electrically isolating the densely-packed transistors in 20nm-and-below memory and logic chip designs with a high-quality dielectric film.

The gaps between these transistors can have aspect ratios of more than 30:1—five times higher than current requirements—and highly-complex profiles. The unique process of Eterna flowable CVD (FCVD) system completely fills these gaps from the bottom up, delivering a dense, carbon-free dielectric film at up to half the cost of spin-on deposition methods that require more equipment and many additional process steps.

Applied's proprietary Eterna FCVD process delivers a liquid-like film that flows freely into virtually any structure shape to provide a bottom up, void-free fill. The system is



Eterna flowable CVD (FCVD) system

BINAY—Lighting Up The Future

BINAY LED

Aviation Obstruction Lights for:

- Industrial Chimneys
- Radio, TV and Structural Towers
- High-rise Buildings & Structures
- Airports and Airfields

Available In Low Intensity and Medium Intensity Versions (as per the International Civil Aviation Organization guidelines), BINAYs PATENTED LED Aviation Lights come with a 5-year Warranty.



BENEFITS

- Rt-and-forget maintenance-free operation
- Long life of 100,000 hours (20 years at 12 hours daily burning)
- Shock-proof and vibration--resistant

THE BINAY LED OBSTRUCTION LIGHT IS UNDER ACCEPTED PATENT, AND AS SUCH IS A PROPRIETARY PRODUCT



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CAUTION NOTICE

Notice is hereby given to all concerned including the Manufacturers, Merchants and Traders dealing inter alia with "LED Aviation Obstruction Lighting Device" Our client, Kumar Krishna Rohatgi of Binay Opto Electronics Private Ltd. is the Patentee of Indian Patent No. 188995 of 9th May 1996 relating to "LED Aviation Obstruction Lighting Device". Therefore, it would not be lawful for any person without the written consent, of our client, to indulge in any act of MAKING, PURCHASING, USING (AS END-USER/CUSTOMER), OFFERING FOR SALE, SELLING OR IMPORTING such device in India during the subsistence of this patent.

It is further made known to all concerned that any unlawful act by any person would not be acceptable to our client, who may decide to take legal action against the infringers for such wrongful act in a Court of Law.

All concerned in the Trade and Commerce and hereby requested to contact the undernoted address or any further details regarding the above and to avoid unnecessary litigation and/or legal proceedings.

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